

ARM PrimeCell

MultiPort Memory controller (PL172)

Release Note

Revision : r2p3

ARM PrimeCell MultiPort Memory Controller (PL172) Release Note

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- The documents number
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General suggestion for additions and improvements are also welcome.

Feedback on the ARM PrimeCell MultiPort Memory controller

If you have any comments or suggestions about this product, please contact your supplier giving:

- The Product name
- A concise explanation of your comments.

Change history

Issue	Date	Change
A	21 December 2001	First release.
B	24 June 2002	Second release for PL172r2p2
C	3 Feb 2003	Third release for PL172 r2p2
D	27 Apr 2004	Release for PL172 r2p3

Product Errata

For Errata relating to this product, please refer to the latest release of the document below.

PL172-DC-11001 ARM PrimeCell MultiPort Memory controller (PL172) Errata Notice (PDF)

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1 PRODUCT DELIVERABLES

1.1 ARM Part Numbers for this product

Table 1.1-1 lists the ARM part numbers for the individual deliverables that form part of the delivery for this ARM PrimeCell peripheral:

Product deliverables for the ARM PrimeCell PL172 peripheral	Document number	Product code
Synthesizable VHDL RTL	N/A	PL172-MN-23001- r2p3-00rel0
Synthesizable Verilog RTL	N/A	PL172-MN-22001- r2p3-00rel0
Synthesis Scripts (Synopsys Design Compiler, Avant! CB18 cell library)	N/A	PL172-MN-01002- r2p3-00rel0
Testbench - Functional Verification, VHDL	N/A	PL172-MN-23010- r2p3-00rel0
Testbench - Integration test, VHDL	N/A	PL172-MN-23012- r2p3-00rel0
Testbench - Functional Verification, Verilog	N/A	PL172-MN-22010- r2p3-00rel0
Testbench - Integration test, Verilog	N/A	PL172-MN-22012- r2p3-00rel0
Test vectors-BusTalk, functional verification	N/A	PL172-VE-01010- r2p3-00rel0
Test vectors-BusTalk, integration test	N/A	PL172-VE-01012- r2p3-00rel0
ARM PrimeCell MultiPort Memory controller (PL172) Technical Reference Manual (PDF)	ARM DDI 0215	PL172-DA-03001- r2p3-00rel0
ARM PrimeCell MultiPort Memory controller (PL172) Design Manual (PDF)	PL172 DDES 0000	PL172-DC-02002- r2p3-00rel0
ARM PrimeCell MultiPort Memory controller (PL172) Integration Manual (PDF)	PL172 INTM 0000	PL172-DC-10002- r2p3-00rel0
ARM PrimeCell MultiPort Memory controller (PL172) Release Note (PDF)	PL172 RELN 0000	PL172-DC-06002- r2p3-00rel0
ARM PrimeCell MultiPort Memory controller (PL172) Errata Notice (PDF)	PL172 DEI 0000	PL172-DC-11001- r2p3-00rel0
ARM PrimeCell MultiPort Memory controller (PL172) Specific Userguide	PL172 USGD 0000	PL172-DC-08001- r2p3-00rel0
Denali Memory Models	N/A	PL172-MN-99001- r2p3-00rel0

The parts with numbers starting PL000 are delivered as a separate bundle PL000-BU-00004

System Common and System Indirection Source Code	N/A	PL000-SW-02001-R9p0-00rel0
System Test Source Code	N/A	PL000-SW-02003-R9p0-00rel0
System Common Files Source Code	N/A	PL000-SW-02004-R9p0-00rel0
System Build Project Files	N/A	PL000-SW-02005-R9p0-00rel0
Interrupt Controller Object Code	N/A	PL001-SW-00000-R9p0-00rel0

Timer Object Code	N/A	PL002-SW-02001-R9p0-00rel0
Software Code Module	N/A	PL172-SW-02001-r2p2-00ltd0
Software Test Report	PL172_TREP_1000	PL172-DC-01001-r2p2-00ltd0
Software API Document	PL172_ISPC_1000	PL172-DC-02001- r2p2-00ltd0
System Structure and Integration Guide	PL000_DII_1000	PL000-DC-02000-R9p0-00rel0
System Common API Documents	PL000_ISPC_1001	PL000-DC-02001-R9p0-00rel0
System Indirection API Documents	PL000_ISPC_1002	PL000-DC-02001-R9p0-00rel0
System Test API Documents	PL000_ISPC_1003	PL000-DC-02003-R9p0-00rel0
System Common Files API Documents	PL000_ISPC_1004	PL000-DC-02004-R9p0-00rel0
Interrupt Controller API Documents	PL001_ISPC_1000	PL001-DC-02001-R9p0-00rel0

Table 1.1-1: ARM part numbers for ARM PrimeCell MPMC PL172

Note Only the required deliverables will be supplied, as per the contractual agreement.

1.2 Reference Cell Library

A reference cell library (Avanti! CB18 0.18µm standard cells) is supplied with this peripheral.

PL000-BU-00005-0.00/Avanti/CB18_2000.3.tar.gz
 PL000-BU-00005-0.00/Avanti/CB18_2000.3.lst

1.3 Common driver bundle

A bundle containing software drivers, common to all PrimeCells is also supplied.

This bundle is part PL000-BU-00004 and is supplied as a single tarred and zipped file,
 PL000-BU-00004-r9p0-00rel0.tgz

2 INSTALLATION

2.1 Introduction

Intellectual Property (IP) deliverables are collectively delivered as a single UNIX zipped tar file which has been security encrypted.

An ARM PrimeCell peripheral may be one of several products delivered as part of this single encrypted file. Each ARM PrimeCell peripheral comprises a set of compressed tar files which correspond to a unique part number quoted in the contractual agreement document. A corresponding list file details the files in each tar file

These installation instructions cover the following platform/operating system:

- UNIX

2.2 Disk space required

This installation will require 400 Mbytes of free disk space.

2.3 Installation procedure

You will have a single tarred, zipped and encrypted file of following format:

E.g.

Company Name-PrimeCell-PL172-Bundled-arm-download-WWWW.tgz.pgp

The installation procedure is summarised below:

2.3.1 Unpacking the shipment

The following steps describe how to unpack the separate products delivered in this shipment.

The first three steps may have already been done, ie the shipment from ARM may have already been decrypted and unzipped. If this is the case then go to step 4, else start with step 1.

1. Relocate the shipment file

Relocate the encrypted file (for example <file>.tgz.des, if DES encryption has been used) to an appropriate location that has sufficient free disk space if PGP encrypted will need the relevant keys.

2. Decrypt file

Follow the accompanying instructions for decryption of the file supplied, to generate a decrypted, compressed, tar file <file>.tgz

3. Unzip file

Execute the UNIX unzip command to obtain the tar file <file>.tar:

```
gunzip <file>.tgz
```

where <file> is the name of the file supplied.

4. Extract tar files

To extract the PrimeCell, execute the UNIX tar extract command:

```
tar -xvf <file>.tar
```

This will generate the following files

ARM_MANIFEST_XXXX.TXT – which lists the PrimeCell files delivered

ARM_DELIVERY_XXXX.TXT – which lists the ARM parts delivered

And will unpack the PrimeCell into the bundle directory PL172-BU-00000-r2p3-00rel0

Beneath this bundle directory are 2 sub-directories

PL172-BU-00000-r2p3-00rel0/PL172_VC contains the PrimeCell development environment

PL172-BU-00000-r2p3-00rel0/PL172_SW contains software for the PrimeCell

The peripheral development environment PL172-BU-00000-r2p3-00rel0/PL172_VC contains two further directory structures, a peripheral specific directory structure and a project global directory structure. The top level directory structures are shown in **Figure 2-1**.

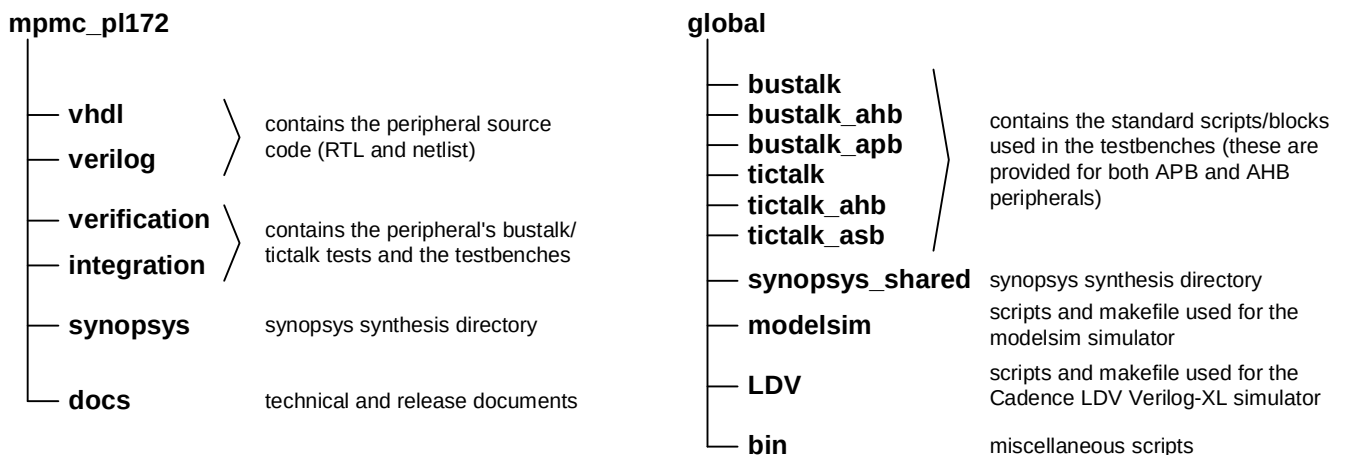


Figure 2-1. Peripheral Development Environment Directory Structure

2.3.2 Installation of the reference cell library

The reference cell library should be copied manually and tar extracted in the directory containing the PrimeCell peripherals. An example is shown below to copy the Avant! CB18 2000.3 cell library to the directory /ARM/PrimeCell_peripherals

```
cd /ARM/PrimeCell_peripherals
cp PL000-BU-00005-0.00/Avanti/CB18_2000.3.tar.gz .
```

Execute the UNIX unzip and tar extract commands as below:

```
gunzip CB18_2000.3.tar.gz
tar -xvf CB18_2000.3.tar
```

2.3.3 Installation of the Common Driver Bundle

The files in the common driver bundle (PL000-BU-00004) unzip and untar into a single directory PrimeCell_gen_sw.

The top level directory structures are shown in **Figure 2-2**.

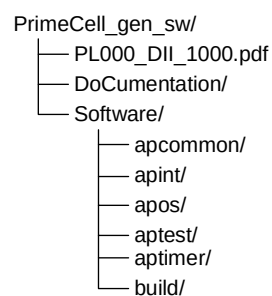


Figure 2-2 Common PrimeCell Driver Directory structure

3 DOCUMENTATION

3.1 Adobe PDF

Documents are supplied as “Adobe PDF” (Portable Document Format) files. These files are readable on most common computer platforms and operating systems using the appropriate file reader. A suitable file reader can be downloaded from the Adobe World Wide Web site at <http://www.adobe.com>. Select “Acrobat Reader” and download the reader for your computer platform/operating system.

3.2 Peripheral Documents Supplied

The following documents are supplied for the ARM PrimeCell MultiPort Memory controller PL172:
The documents files are located in the `mpmc_pl172/docs/` directory.

ARM PrimeCell MultiPort Memory controller PL172 Technical Reference Manual ARM DDI 0215

The technical reference manual provides details for programming and interfacing to this ARM PrimeCell peripheral block.

ARM PrimeCell MultiPort Memory controller PL172 Integration Manual PL172 INTM 0000

The integration manual provides information required to integrate this ARM PrimeCell block within a typical industry standard ASIC design flow.

ARM PrimeCell MultiPort Memory controller PL172 Design Manual PL172 DDES 0000

The design manual describes the implementation details of this ARM PrimeCell block and the testbenches and test programs used for functional verification.

ARM PrimeCell MultiPort Memory controller PL172 Errata Notice PL172 DEI 0000

The errata notice describes details of any known issues with this ARM PrimeCell block and any workarounds for these issues.

ARM PrimeCell MultiPort Memory controller PL172 User Guide PL172 USGD 0000

The User Guide describes the instructions on performing simulations and synthesis for this ARM PrimeCell block.

4 TRIAL TESTING

4.1 Product Maturity

This product is newly developed IP and is initially released with Early Access status.

Early Access release status has a particular meaning to ARM of which the recipient must be aware. A product so designated has satisfactorily achieved all criteria for its promotion to Full Release status. It may be delivered in accordance with the contract and be expected to perform as described in the data-sheet. However, there remains some elements of uncertainty, which cannot finally be validated until the product has been successfully deployed by customers. Accordingly, the recipient of a product with Early Access-Release status, may be directly contributing to the final stage of approval of that product.

In due course, the Early Access-Release will be replaced by the Full-Released version or where no changes have been required to the product's deliverables a simple notification will be given that the product has now been given Full Release status. This release note will not necessarily be updated to indicate a move to Full Release. It should be noted that Support for the Early Access release of the product will only be provided by ARM to a recipient who has a current support and maintenance contract for the product.

4.2 Technical data summary

The following is a summary of the key technical points from testing of the ARM PrimeCell MultiPort Memory controller PL172.

	VHDL RTL	Verilog RTL
Cell library	Avant! Passport 2000.3 CB18 v1.0 (0.18 μ m)	Avant! Passport 2000.3 CB18 v1.0 (0.18 μ m)
Total cell area, without scan, NAND-2 equivalents	78,800	79,399
Net interconnect area, without scan, NAND-2 equivalents	32,142	32,743
Total cell area, scan inserted, NAND-2 equivalents	98,128	98,104
Net interconnect area, scan inserted, NAND-2 equivalents	39,956	39,529
Clock frequency used	151 MHz	151MHz
Scan insertion figure % (number of 'D' flip-flops replaced)	100% (6366)	100% (6366)
Estimated scan fault coverage, collapsed.	99.99%	99.97%
Synthesis and scan test tools and version	Synopsys Design Compiler and Test Compiler 2002.05-SP1-3	Synopsys Design Compiler and Test Compiler 2002.05-SP1-3
Denali memory models version	3.000-0031	3.000-0031
Simulation tools and versions	Model Technology Inc, ModelSim v5.5d	Model Technology Inc, ModelSim v5.5d Cadence Verilog-XL 3.2
Code coverage tools and version	Transeda Navigator 6.2.2_modelsim5.5d	Transeda Navigator 6.2.2_modelsim5.5d
Global Peripheral Development Environment Version	Release REL1v7	

4.3 BusTalk Functional block verification test scenarios

Functional block verification has been performed with the following scenarios using the BusTalk testbench with free-running clocks.

4.3.1 Synchronous clocks

MPMCCLK/PCLK	Simulations
151 MHz	VHDL and Verilog, RTL and gate-level min/max SDF

For further details of functional verification testing, refer to the ARM PrimeCell MultiPort Memory controller (PL172) Design Manual.

5 DIFFERENCES FROM PREVIOUS REVISIONS

5.1 Differences between r2p2-00ltd0 and r2p2-00ltd1

There are no functional changes between these 2 versions.

The testbench which was shipped with version r2p2-00ltd0 was incorrect. r2p2-00ltd1 contains the correct version.

The Integration Manual has been extended.

5.2 Differences between r2p2-00ltd1 and r2p3-00rel0

The following defects have been corrected in this release. Refer to the Errata document for details.

319664: (Category 1) Some SDRAM writes missed and not written to memory

284509: (Category 2) CKE signal is low at power-up.

314789: (Category 3): Synthesis should be HCLK:MPMCCLK = 1:1